

MIRI, Malaysia

WMO: 964490

Lat: 4.322N

Lon: 113.987E

Elev: 59

StdP: 14.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	72.8	73.2	69.5	108.7	79.3	70.3	111.7	78.7	16.4	84.0	14.5	84.1	3.8	100	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.4	91.3	79.9	90.0	79.7	89.7	79.6	82.3	88.1	81.6	87.5	81.0	87.0	7.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	159.8	86.1	79.9	155.7	85.3	79.2	151.8	84.4	46.3	88.1	45.3	88.1	44.8	87.7	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	11.9	10.7	69.6	93.6	3.3	1.1	67.2	94.4	65.3	95.1	63.5	95.7	61.1	96.5
			69.1	84.4	1.7	1.7	67.9	85.7	66.9	86.7	65.9	87.7	64.6	89.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	80.4	80.6	81.3	82.0	82.4	82.0	81.7	81.8	81.6	81.2	81.1	80.9
	DBStd	1.70	1.79	1.70	1.59	1.47	1.64	1.58	1.62	1.69	1.52	1.53	1.33	1.58
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11471	942	856	971	961	1005	960	982	985	949	967	934	959
	CDD65	5996	477	436	506	511	540	510	517	520	499	502	484	494
	CDH74	60316	4169	3861	4923	5449	5973	5429	5407	5472	5200	5034	4752	4647
	CDH80	20255	1135	1037	1547	1928	2273	1994	1989	1999	1832	1651	1491	1379
Wind	WSAvg	4.5	4.7	4.8	4.5	4.2	4.3	4.3	4.4	4.6	4.6	4.6	4.5	4.6
Precipitation	PrecAvg	112.0	12.8	6.3	6.2	6.8	7.6	8.3	8.0	8.7	8.8	11.9	12.4	13.9
	PrecMax	135.9	68.3	14.9	14.4	15.7	14.8	25.6	22.2	14.6	16.9	19.6	27.7	27.3
	PrecMin	81.5	0.3	0.1	0.3	0.2	1.1	2.4	1.5	1.7	2.3	4.6	2.1	5.0
	PrecStd	14.6	11.2	4.3	3.5	3.1	3.7	4.2	4.0	3.6	3.8	4.0	4.5	5.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	89.8	90.8	91.3	91.9	91.7	91.5	91.6	91.2	90.1	89.9	89.9
		MCWB	79.6	79.0	79.3	80.8	80.6	79.6	79.1	78.9	79.8	79.2	79.8	79.8
	2%	DB	88.0	88.0	89.2	89.8	90.6	90.0	89.9	89.9	89.7	89.3	88.2	88.3
		MCWB	79.2	79.0	79.3	80.2	80.5	79.5	79.4	79.3	79.3	79.3	79.5	79.5
	5%	DB	86.5	86.6	87.8	88.9	89.6	89.5	89.3	89.2	88.7	88.0	87.7	87.6
		MCWB	78.7	78.6	79.3	80.0	80.2	79.4	79.3	79.3	79.2	79.1	79.4	79.3
	10%	DB	85.6	85.7	86.5	87.8	88.5	88.1	88.0	87.9	87.7	87.0	86.4	86.2
		MCWB	78.3	78.3	78.8	79.8	80.0	79.3	79.1	79.0	79.1	78.8	78.9	78.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	81.0	81.7	82.6	83.6	82.1	82.3	81.9	82.5	81.7	81.8	81.6
		MCDB	86.7	86.5	87.3	88.5	88.6	88.6	88.6	88.2	88.1	87.3	87.2	87.1
	2%	WB	80.1	80.1	80.7	81.6	82.1	81.1	81.0	81.0	80.9	80.5	80.7	80.6
		MCDB	85.6	85.5	86.6	87.4	87.8	87.6	87.6	87.3	87.1	86.7	86.2	86.3
	5%	WB	79.4	79.3	80.0	80.8	81.3	80.4	80.2	80.2	80.1	79.8	79.9	79.8
		MCDB	84.6	84.5	85.5	86.5	87.1	86.9	86.7	86.4	86.2	85.7	85.3	85.2
	10%	WB	78.9	78.8	79.3	80.1	80.5	79.8	79.6	79.6	79.5	79.2	79.3	79.2
		MCDB	83.7	83.8	84.5	85.5	86.5	85.9	85.7	85.5	85.3	84.7	84.4	84.1
Mean Daily Temperature Range	5% DB	MDBR	11.1	10.9	11.8	12.0	12.4	12.5	12.8	12.6	12.3	11.8	11.6	11.3
		MCDBR	12.3	12.2	12.9	13.2	13.6	14.0	14.1	14.0	13.7	13.2	12.7	12.8
		MCWBR	5.1	4.9	5.3	5.5	5.7	5.8	5.9	5.9	5.9	5.7	5.6	5.3
	5% WB	MCDBR	11.5	11.2	11.9	12.2	12.7	12.9	13.0	12.9	12.8	12.3	11.9	11.8
		MCWBR	5.3	5.1	5.5	5.6	6.0	5.8	6.1	5.9	6.0	5.7	5.6	5.3
Clear-Sky Solar Irradiance	taub	0.423	0.435	0.434	0.423	0.408	0.414	0.415	0.460	0.458	0.465	0.415	0.418	
	taud	2.471	2.416	2.416	2.446	2.501	2.489	2.483	2.331	2.333	2.321	2.481	2.490	
	Ebn at Noon	285	284	283	281	279	273	274	267	272	272	286	285	
	Edh at Noon	36	39	39	37	34	34	34	41	42	42	36	35	
All-Sky Solar Radiation	RadAvg	1495	1673	1800	1802	1683	1597	1615	1644	1614	1521	1529	1482	
	RadStd	139	118	87	69	66	67	79	55	99	93	83	94	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.49	+0.41	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page